

Mame Diarra Dieng

AI & Big Data Engineering Student | Dakar, Senegal
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PROFILE

Final-year (DIC3) engineering student in AI & Big Data at ESP Dakar. I design and deploy end-to-end data and AI systems, from ingestion to production, and I build software products for the Senegalese market. Currently AI Workflow & Enablement Engineer at CONCREE, supporting every department in adopting AI.

EDUCATION

École Supérieure Polytechnique de Dakar (ESP – UCAD)

Engineering Degree in Artificial Intelligence & Big Data

2024 – Present (expected 2027)

Dakar, Senegal

- Coursework: Machine Learning, Big Data, Distributed Systems, Virtualization Infrastructure, Parallel Computing (GPU/CUDA), Business Intelligence, NoSQL, Cloud Computing, Probability & Statistics

École Supérieure Polytechnique de Dakar (ESP – UCAD)

University Diploma of Technology (DUT) in Computer Science

2022 – 2024

Dakar, Senegal

HONORS & COMPETITIONS

- **Sococim Innovation Challenge 2026: Finalist.** Team selected among 400 applications, 20 teams retained (top 5%) for the final phase.
- **Finalist, Polytechnic Alumni Forum – New Deal Technologique (2024):** AgriGrenier, an IoT platform for monitoring agricultural storage.

PROFESSIONAL EXPERIENCE

CONCREE – AI Workflow & Enablement Engineer (6-month internship)

Reporting to the General Management

Sept. 2026 – Feb. 2027

Dakar, Senegal

- Support every department (marketing, communications, entrepreneurship support programs, business development, operations, product/tech, management) in integrating AI into their business workflows
- Build a library of reusable AI Skills, deployable in the internal Kangam platform and in ChatGPT / Claude-type environments
- Follow a structured approach: workflow mapping, opportunity identification, prototyping, deployment, team training and impact measurement
- Run action-oriented technology watch: agents, MCP, RAG, tool calling, automation & no-code

SONATEL (Orange Senegal) – Data Engineering Intern

End-of-DUT internship

May – Aug. 2024

Dakar, Senegal

- Designed and deployed a pipeline processing 1M+ records per day from multiple sources using the Elastic Stack (Elasticsearch, Logstash, Kibana)
- Wrote advanced Logstash configurations that cut manual preprocessing by 40%
- Built Kibana dashboards for real-time KPI tracking, reducing analysis time by 30%
- Documented the full pipeline for internal knowledge transfer

TECHNICAL PROJECTS

AgriGrenier – IoT Agricultural Monitoring Platform

Arduino, Flask, Supabase, Twilio, n8n

2024

github.com/diarraaaa/AgriGrenierIoT

- Arduino sensors measuring temperature, humidity and storage conditions in real time
- Flask REST API receiving and processing sensor data, with persistence in Supabase
- Automated SMS alerts via Twilio when a critical threshold is crossed, orchestrated with n8n

Book Recommendation Engine

March 2026

Python, Flask, Pandas, TF-IDF, Sentence Transformers, SQLite

- Collected and cleaned a dataset of 50,000+ books (normalization, missing values)
- Built semantic models with TF-IDF and Sentence Transformers; hybrid engine combining content-based filtering and user interactions
- Flask REST API, SQLite persistence, HTML/CSS/JavaScript web interface

ENTREPRENEURSHIP & PRODUCTS

Dalbix – CTO, Short-Term Rental Platform

2026 – Present

Django REST, Next.js, PostgreSQL, Firebase, Redis, Celery, Docker

dalbix.net

- Short-term rental platform with owner, guest and admin roles: property listings, bookings with counter-offers, payments, maintenance requests and real-time chat
- Django REST API (business logic, permissions) with a Next.js / React / Tailwind frontend; PostgreSQL in production, SQLite in development
- Authentication with Firebase ID tokens verified server-side; user roles resolved from Firestore instead of being trusted from the client token
- Redis for shared cache, sessions and rate limiting across multiple Gunicorn workers; Celery for asynchronous email with automatic retries
- Full local stack (backend, Redis, Celery worker) run with Docker Compose

Freelance Web Developer – Univers Éléphant

2026 – Present

React, Vite, Supabase, n8n, Vercel

universelegant.com

- Website and customer area for a made-to-measure Senegalese traditional clothing brand, built around a custom design system (Bodoni Moda / Jost, warm palette, angular style)
- Supabase backend (accounts, measurements, orders) secured with row-level security (RLS) policies
- Order flow without online payment: generates a pre-filled WhatsApp message
- AI chat assistant connected to an n8n workflow exposing product catalog, order tracking and ticket creation tools, querying the Supabase database directly
- Continuous deployment on Vercel on every push to the main branch

TECHNICAL SKILLS

Languages: Python, PHP, SQL

Frameworks & Backend: Flask, Django REST, React

Data Science & AI: Pandas, Scikit-learn, Sentence Transformers

Applied AI & LLMs: RAG, AI agents, MCP & tool calling

Big Data & Engineering: Elastic Stack (ELK), Hadoop, ETL / Business Intelligence

Databases: PostgreSQL, Supabase, MongoDB

DevOps & Cloud: Docker, Vercel, Git / CI-CD

IoT & Automation: Arduino, n8n, Twilio

LANGUAGES & INTERESTS

Languages: French (native), English (professional), Arabic (reading), Wolof (native)

Interests: Tennis, reading, cinema